

Database Management Systems

Entity-Relationship Model

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PES UNIVERSITY

Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of "Database System Concepts", Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems", Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor



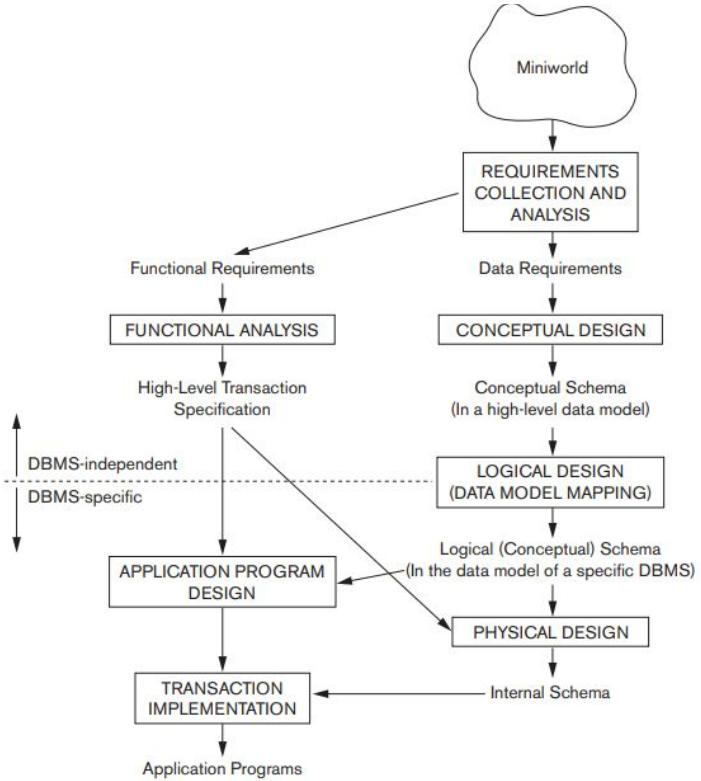
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Database Design Phases

Design Phases:

- Characterizing User Data Needs
- Choosing a Data Model and Translating Requirements
- Review and Refinement of Schema
- Functional Requirements Specification
- Logical-Design Phase
- Physical-Design Phase



E-R model

- E-R diagrams provide a graphical representation of the database's logical structure. They use symbols to depict entities, relationships, and attributes, making complex relationships easier to understand.
- E-R diagrams are valued for their simplicity and clarity, making them a widely used tool in database design.
- By mapping out the entities, attributes, and relationships before implementation, E-R Diagrams help in identifying potential issues, redundancies, or inconsistencies in the database design. This leads to more efficient and optimized database structures, reducing the need for costly modifications later.
- E-R Diagrams serve as a comprehensive documentation tool that records the structure of the database.

E-R model - Description of Company Database

- The company is organized into DEPARTMENTs.
- Each department has a name, number and an employee who manages the department.
- Keep track of the start date of the department manager.
- A department may have several locations.
- Each department controls number of PROJECTs.
- Each project has a unique name and number and is located at a single location.

E-R model - Description of Company Database

- The database will store each EMPLOYEE's social security number, address, salary, sex (gender) and birthdate.
- Each employee works for one department but may work on several projects.
- The DB will keep track of the number of hours per week that an employee currently works on each project.
- It is required to keep track of the direct supervisor of each employee.
- Each employee may have a number of DEPENDENTs.For each dependent, the DB keeps a record of name, sex, birthdate, and relationship to the employee.

Entity

- Entity is a basic concept for the ER model.
- Entities are specific things or objects in the mini-world that are represented in the database.
- Entity can have physical existence or conceptual existence
- Physical (car, person, employee, house)
- Conceptual (a company, a job, a university course, a project)
- Each entity has attributes
- Entity Set is a collection of similar entities

Entity

- **Unique Identification of entity:**
  - Unique property values identify each entity.
  - For instance, a student’s “student_id" value like PES001 makes them distinct.
  - Courses can also be entities in a university setting.
  - A "course_id" attribute uniquely identifies each course.

- **Concrete (Physical) and Abstract (Conceptual) Entities:**
  - Entities can be concrete (like people or books) or abstract (like courses or reservations).
  - Both types serve as distinct objects in the system.

Entity Sets

- **Entity Set (or Entity Collection):**
  - An entity set is a group of entities of the same kind sharing common attributes or properties.
  - For instance, "instructor" represents all university instructors, and "student" represents all university students.

E1 is an entity having Entity Type Student and the set of all students is called Entity Set.

There are two types of Entities:

Strong Entity and Weak Entity

Entity Sets

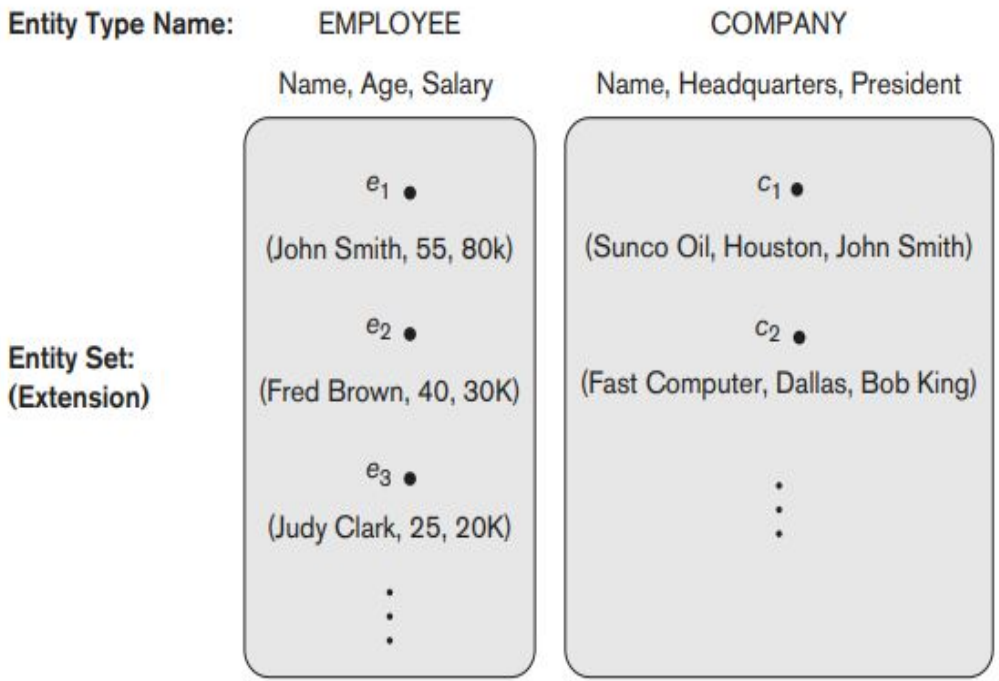


Figure 3.6
Two entity types, EMPLOYEE and COMPANY, and some member entities of each.

Attributes

- Entities are represented by specific attributes.
- Attributes are qualities or characteristics of each entity.
- They signify the type of information stored for each entity.
- Every entity possesses its own set of values for attributes.
- Example: Instructors have unique values for attributes like ID, name, dept name, and salary.

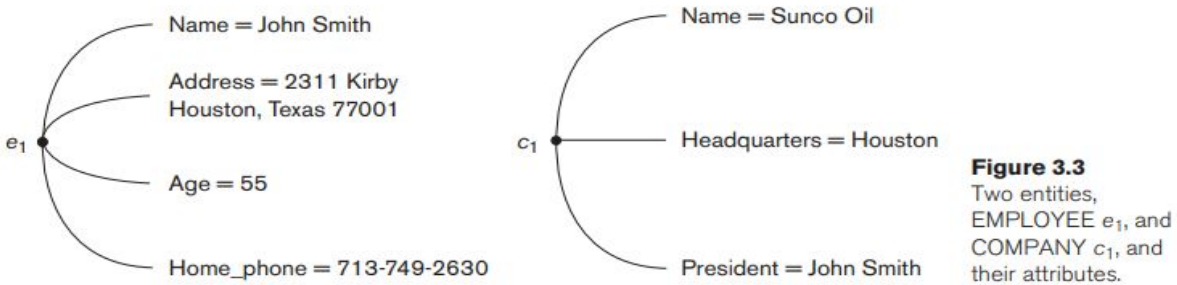
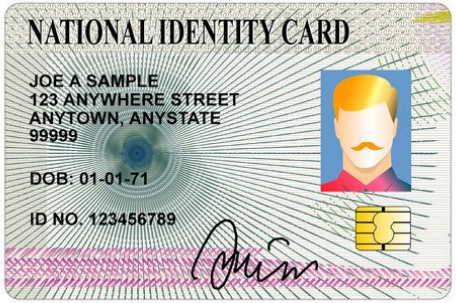


Figure 3.3
Two entities, EMPLOYEE e_1, and COMPANY c_1, and their attributes.

Attributes

- An entity's attributes provide distinct values for identification.
  - For instructors, the ID attribute stands out as a unique identifier.
- It prevents confusion when multiple instructors share the s name.
- How do we decide on the unique identifier for the case of Instructors?
  - Government-issued IDs were historically used, but they raise security and privacy concerns.
  - Enterprises opt to establish their own identifiers to ensure security and privacy.

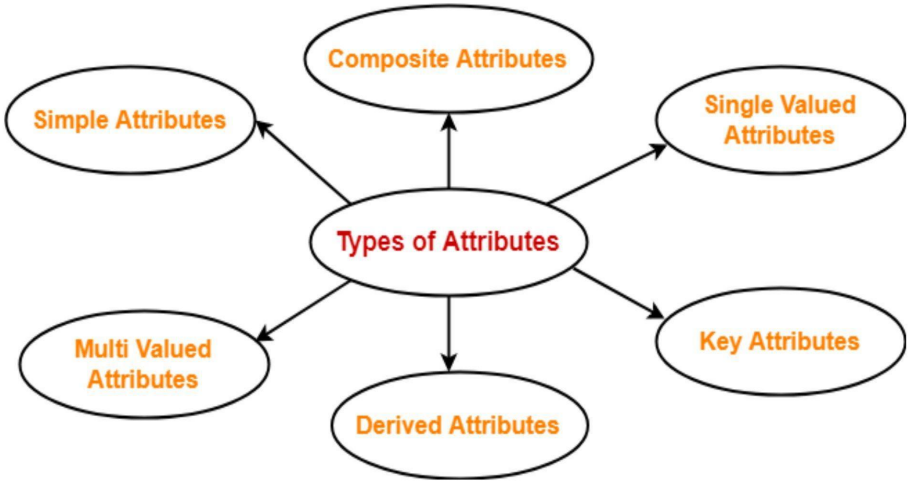


Value Set (Domain)

- For each attribute there are a set of permitted values that can be taken which is called DOMAIN or VALUE SET of that attribute.
  - If the range of ages allowed for employees is between 16 and 70, we can specify the value set of the Age attribute of EMPLOYEE to be the set of integer numbers between 16 and 70.
  - We can specify the value set for the Name attribute to be the set of strings of alphabetic characters separated by blank characters, and so on.
- Mathematically, an attribute A of entity set E whose value set is V can be defined as a function from E to the power set6 P(V) of V: A : E $\rightarrow$ P(V)

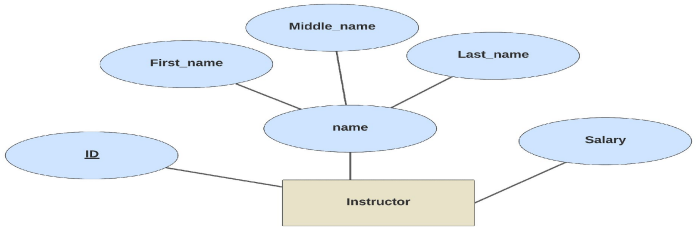
Types of Attributes

- An attribute used in E-R model, can be characterized as:
  - **Simple / Composite attribute**
  - **Single-valued / Multivalued attributes**
  - **Stored / Derived attributes**
  - **Complex Attribute**
  - **Key Attribute**



Simple and Composite Attribute

- Formally saying the attributes that cannot be further subdivided into components is a **simple attribute**
  - Ex: The ID of the student or Instructor, Roll number of the student, phone number, email id, etc.
- On the contrary those attributes that can be split into components are called a **composite attribute**
  - Ex: The address can be further split into house number, street number, city, state, country, and pin code,
  - The name can also be split into first name middle name, and last name.



Composite Attributes

- Let us consider the university database. Like any other organization, the management would keep a record of all the addresses of the student, instructors and the staff
- Now address would contain the entire street, city, state, and postal_code information.
- Let us say the address of the student is :

  "123 Main Street, Yeshwanthpur, Bangalore, 560063"

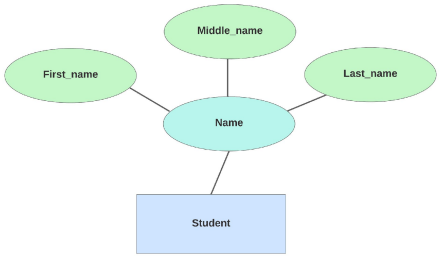
  Now this is how all the information is stored in one attribute.
  - Let us say we would want to group students through the postal code for allotting the busses. How do we do it?
  - Currently the postal code is not an attribute alone or stored specifically as an attribute so we cannot get the postal code alone as it is a part of the address

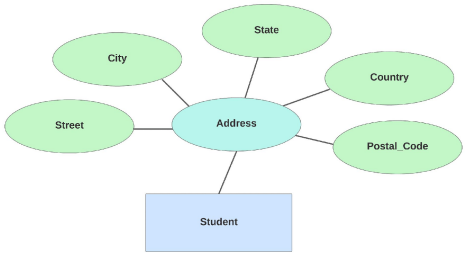
Composite Attributes

- Thus just having the address stored as one attribute is not always appropriate, especially when we would want to retrieve students on a part of the address.
- For solving this issue, we try to store these types of attributes by dividing the attribute into subparts.
- These attributes which can be divided into subparts are called '**Composite attributes**'.
- Using composite attributes in a design schema is a good choice if a user will wish to refer to an entire attribute on some occasions, and to only a component of the attribute on other occasions.

Representation of the Composite attribute

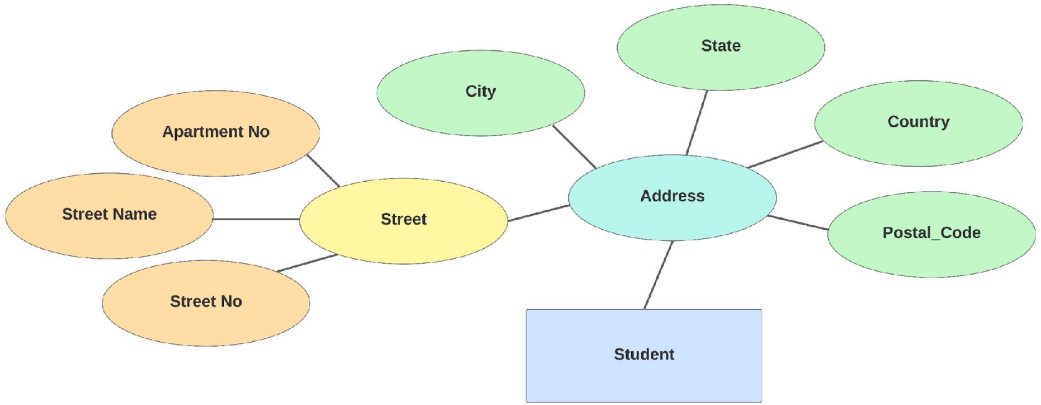
- Let us consider the examples discussed above, and how are they represented it in the E-R diagram
  - Name could be subdivided into First Name, Middle Name, Last Name
  - The address can be further split into street, city, country, and pin code





Representation of the Composite attribute

- Composite attributes may appear as a hierarchy.
- In the composite attribute address, its component attribute street can be further divided into street number, street name, and apartment number.



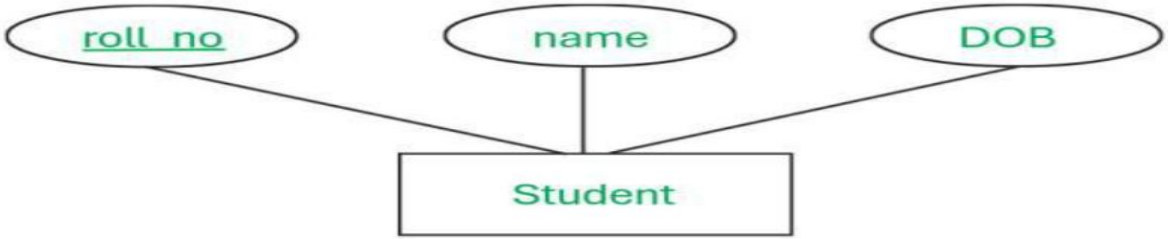
Multivalued Attributes

- Suppose the university has decided to add the phone number to the instructor relation.
- An instructor may have Zero, One, or Several phone numbers, and different instructors may have different number of phone numbers.
- Same is the case with the instructor's mail_id.
- Till now all the attributes we have studied were having only a single value and not multiple values. So how do we deal with these type of situation ?

- These types of attributes which can have multiple values are called as **multivalued Attributes**
- Examples for the same are :
  - Instructor Phone numbers
  - Instructors mail_id

Single-valued and Multivalued Attributes

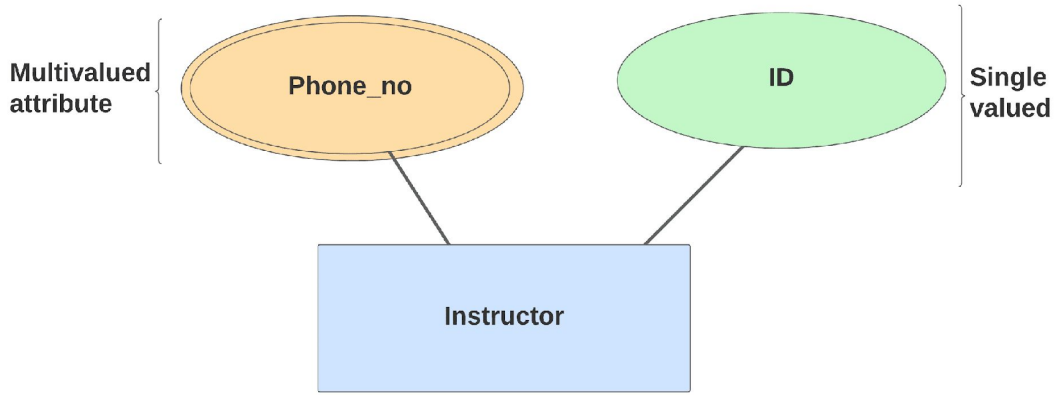
- The attribute which takes up only a single value for each entity instance is a **single-valued attribute**.
  - Example: The name of the instructor, age of the student, etc.
- The attribute which takes up more than a single value for each entity instance is a **multi-valued attribute**.
  - Example: Phone number of the instructor, Mail id, etc.



Single-Valued

Representation of Multivalued Attributes

- The multivalued attribute is represented in the E-R diagram as shown below:
- The multivalued attribute is represented by a double ellipse.
- Considering the instructor's Phone_no as a multivalued attribute and ID as an single-valued attribute:



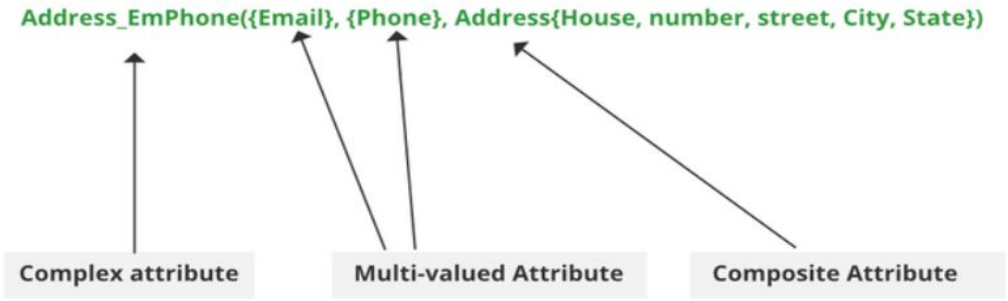
Complex Attributes

- Those attributes, which can be formed by the nesting of composite and multi-valued attributes, are called "**Complex Attributes**".
- These attributes are rarely used in DBMS.

- Let us take an example of the same.
  - Let us consider that the student relation contains an attribute called PreviousDegrees that stores all the previous education qualifications of the students
  - PreviousDegrees of a STUDENT is a composite multi-valued attribute denoted by {PreviousDegrees (College, Year, Degree, Field)}
  - Multiple PreviousDegrees values can exist. Each has four subcomponent attributes: College, Year, Degree, Field
  - Therefore it is a complex attribute

Question

Given ,an attribute Address_EmPhone consisting of email, phone and Address. Can you identify which are the different attribute types that make up this attribute.



Complex attributes

Derived Attributes

- Consider the following schema for the Instructor relation
  - **Instructor** (ID, Name, Department, Date_of_Birth, Age)

- Can you identify any problem in the above schema?
- If you have noticed, we know that the age can be calculated from the Date of birth, i.e. we can take the system's current date and the Date of birth of the instructor and we can find the age.

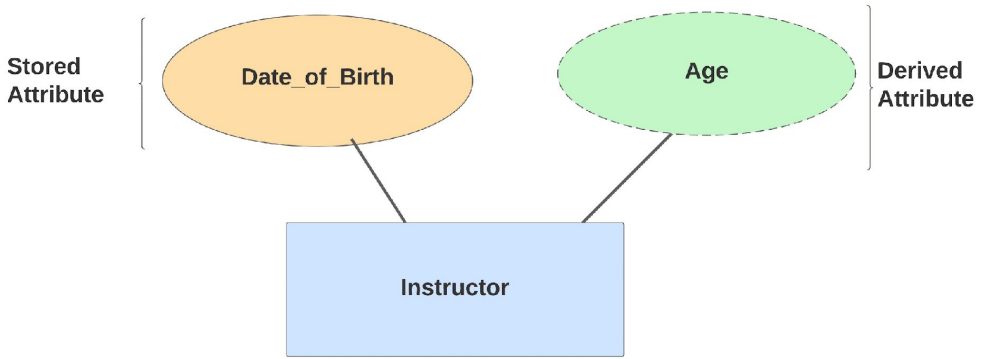
- So there is no need to store this information in the actual database and can be calculated.

Stored and Derived Attributes

- So as shown above, the value of the age attribute can be derived from the date of Birth.
- Therefore the Age attribute is called a Derived Attribute
- And the Date_of_Birth attribute is called as a stored attribute.
- The **stored attribute** are those attributes that are actually stored in the database.
  - Example: Date_of_Birth is a stored attribute
- An attribute that can be derived from other attributes is **derived attributes**.
  - Example: the Age attribute of the instructor is a derived attribute

Representation of Derived Attribute

- Considering the above example of the instructor relation, Date_of_Birth is a stored attribute and Age is an derived attribute
- In the below E-R Diagram, Age (derived attribute) is represented by a dashed oval.
- The E-R representation would be:



Key Attribute

- An attribute of an entity type for which each entity must have a unique value is called a key attribute of the entity type.

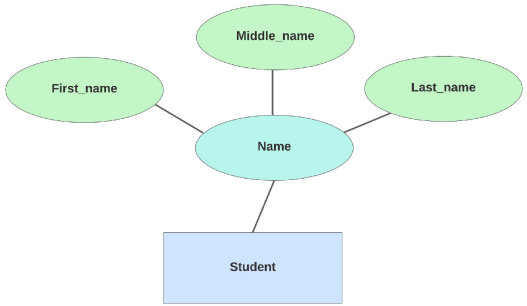
  Example: SSN of Employee, SRN of a Student.

Key Attribute

- A key attribute may be composite.
- VehicleTagNumber is a key of the CAR entity type with components (Number, State).
- An entity type may have more than one key.
- The CAR entity type may have two keys:
- VehicleIdentificationNumber (popularly called VIN)
- VehicleTagNumber (Number, State), aka license plate number.
- Each key is underlined (Note: this is different from the relational schema where only one “primary key is underlined).

Null Values

- Let us consider the diagram representation shown:
- What if the student doesn't have a middle name?



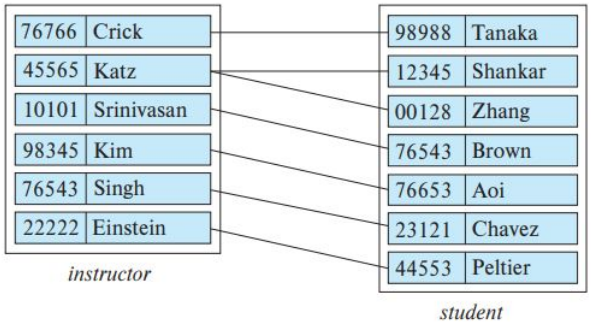
- In such situations the value of the middle name would be set to NULL

Null Values

- An attribute takes a null value when an entity does not have a value for it.
- So what does it mean if an attribute is NULL?
- The Null value may indicate "**not applicable**"—that is, the value does not exist for the entity.
- Null can also designate that an attribute value is **unknown**. An unknown value may be either:
  - missing (the value does exist, but we do not have that information)
  - not known (we do not know whether or not the value actually exists).
- A null value for the apartment number attribute could mean:
  - Address does not include an apartment number (not applicable),
  - That an apartment number exists but we do not know what it is (missing),
  - We do not know whether or not an apartment number is part of the instructor's address (unknown).

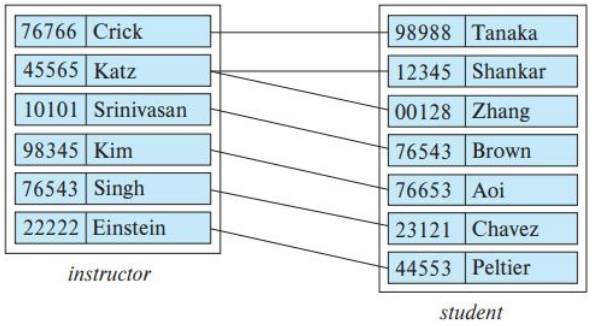
Relationship and Relationship sets

- **What is a Relationship:**
  - A relationship signifies connections between entities.
  - Example: "advisor" links instructor Katz to student Shankar.
- **Relationship Set:**
  - A relationship set contains relationships of the same type.
  - Example: "advisor" relationship set links students and their advisors.
- **Entity Sets and Relationships:**
  - Consider "instructor" and "student" entity sets.
  - "advisor" relationship set signifies student-adviso connections.



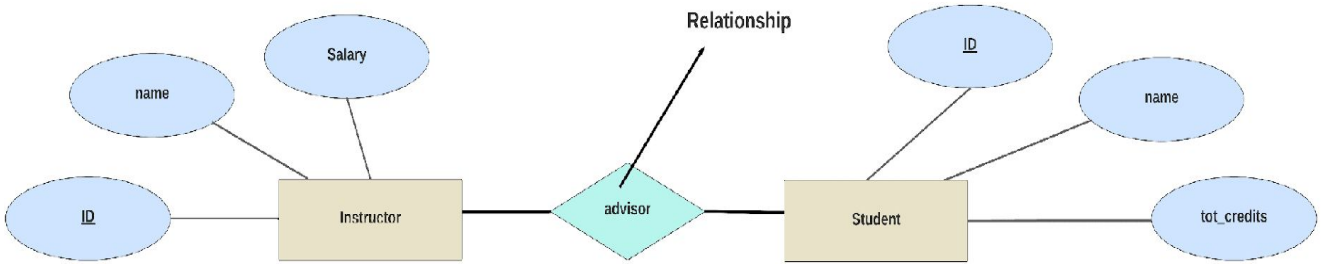
Relationship and Relationship sets

- **Relationship Instance:**
  - A relationship instance in an E-R schema represents an association between the named entities in the real-world enterprise that is being modeled.
  - As an illustration, the individual instructor entity Katz, who has instructor ID 45565, and the student entity Shankar, who has student ID 12345, participate in a relationship instance of advisor.
  - This relationship instance represents that in the university, the instructor Katz is advising student Shankar



Relationship and Relationship sets

- **Representation in E-R Diagram:**
  - In an E-R diagram, relationships are depicted by diamonds.
  - Lines connect the diamond to relevant entity sets (rectangles).
- For example let us try and represent the advisor relation



Recursive Relationships

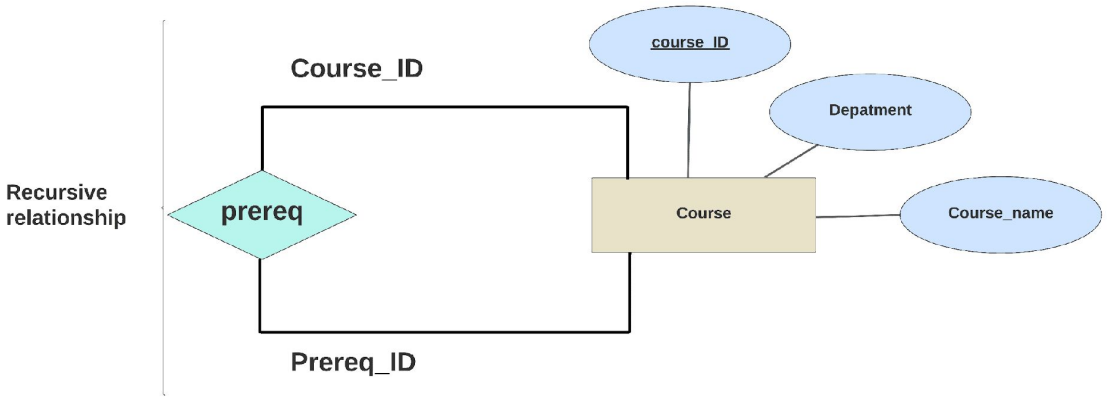
- The function that an entity plays in a relationship is called that entity's **role**.
- Since entity sets participating in a relationship set are generally distinct, roles are implicit and are not usually specified.
- However, they are useful when the meaning of a relationship needs clarification. Such is the case when the entity sets of a relationship set are not distinct; that is, the same entity set participates in a relationship set more than once, in different roles.
- In this type of relationship set, sometimes called a **recursive relationship set**, explicit role names are necessary to specify how an entity participates in a relationship instance.

Recursive Relationships

- Let us try and understand it with the help of an example
- consider the entity-set course that records information about all the courses offered in the university. Now some courses that are offered by the university have some prerequisite courses which are needed to be completed
- To depict the situation where one course (C2) is a prerequisite for another course (C1) we have a relationship set prereq that is modeled by ordered pairs of course entities.
- So how do we indicate this in the ER diagram?

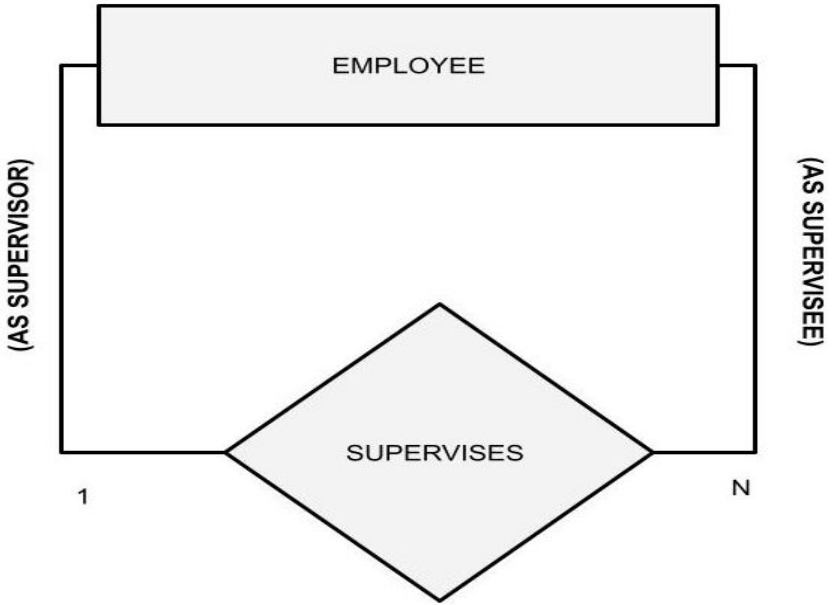
Recursive Relationships

- We indicate roles in E-R diagrams by labeling the lines that connect diamonds to rectangles.
- The diagram below shows the role indicators course id and prereq id between the course entity set and the prereq relationship set.



Recursive Relationship - Employee - Supervisor Example

Can you think of another example of a recursive-relationship?



Relationship Set and it's degree

- The relationship sets advisor is an example of a **binary relationship set**—that is, one that involves two entity sets.
- Most of the relationship sets in a database system are binary.
- Occasionally, however, relationship sets could even involve more than two entity sets. The number of entity sets that participate in a relationship set is the **degree of the relationship set**.
  - A binary relationship set is of degree 2;
  - A ternary relationship set is of degree 3.

Relationship Set Example

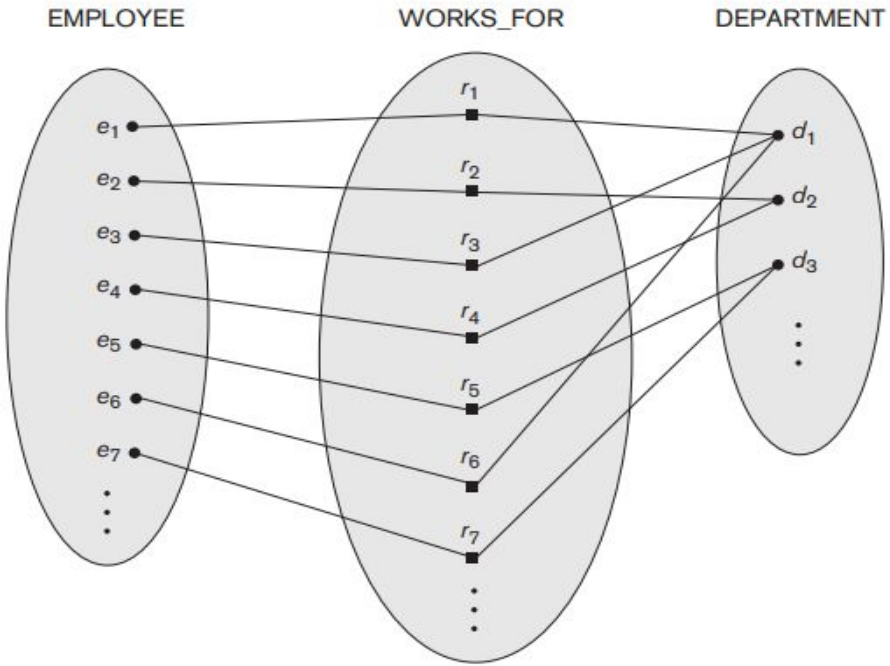


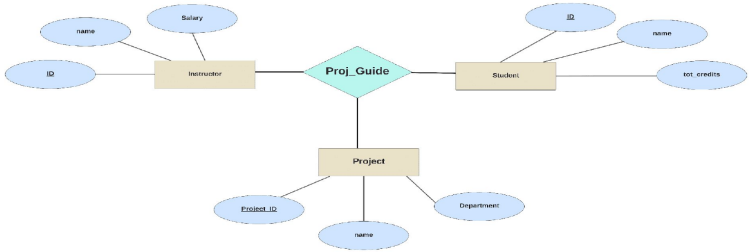
Figure 3.9
Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

Ternary Relationships

- Let us consider an example, suppose that we have an entity set project that represents all the research projects carried out in the university.
- Consider the entity sets instructor, student, and project. Each project can have multiple associated students and multiple associated instructors. Furthermore, each student working on a project must have an associated instructor who guides the student on the project.
- Suppose we were to represent the information of which instructor is guiding which student for which project
- How do we represent this or model this in an ER diagram?

Ternary Relationship Example

- To represent the above situation, we would have to relate the three entity sets through a ternary relationship set proj_guide, This ternary relation relates entity sets instructor, student, and project.
- An instance of projguide indicates that a particular student is guided by a particular instructor on a particular project.
- Note that a student could have different instructors as guides for different projects, which cannot be captured by a binary relationship between students and instructors.

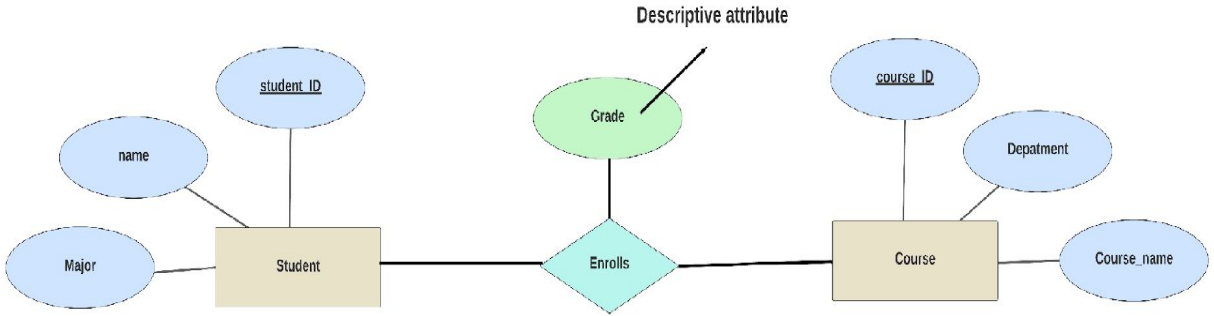


Relationship Attribute

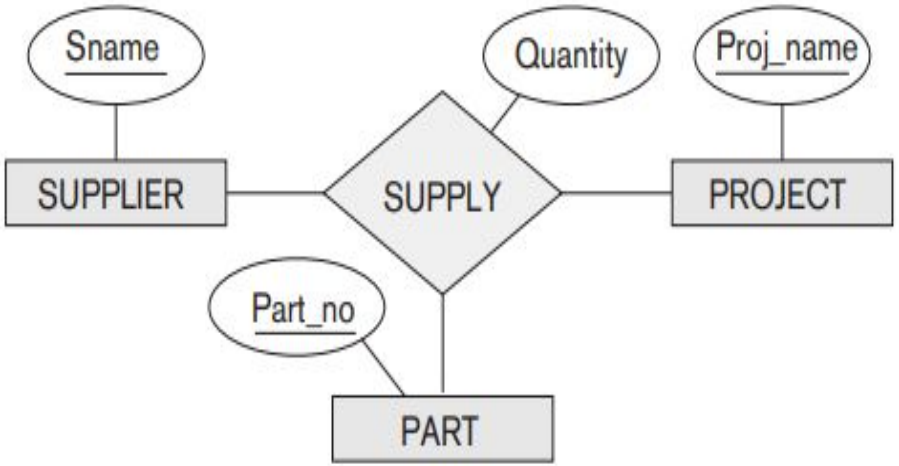
- Let us now consider the scenario that every student enrolls for certain courses.
- Now as we know along with what each student enrolls for which course, the university must also keep track of the grades that the student would score for that particular course.
- Now where do we place this grade attribute? In the Student entity or the Courses entity?
- Actually if we observe this attribute is associated with the relation ‘enroll’ i.e. only if the student enrolls for the particular course, he can get a grade. Such type of attributes are called **Relationship attributes**
- So it wouldn’t be correct to place it in any of the entities
- Then where do we place them?

Relationship Attribute

- A relationship may also have attributes called descriptive attributes.
- An attribute of a relationship set is represented in an E-R diagram by an oval connected to the relationship's diamond
- The below diagram shows the discussed example in the form of an ER diagram



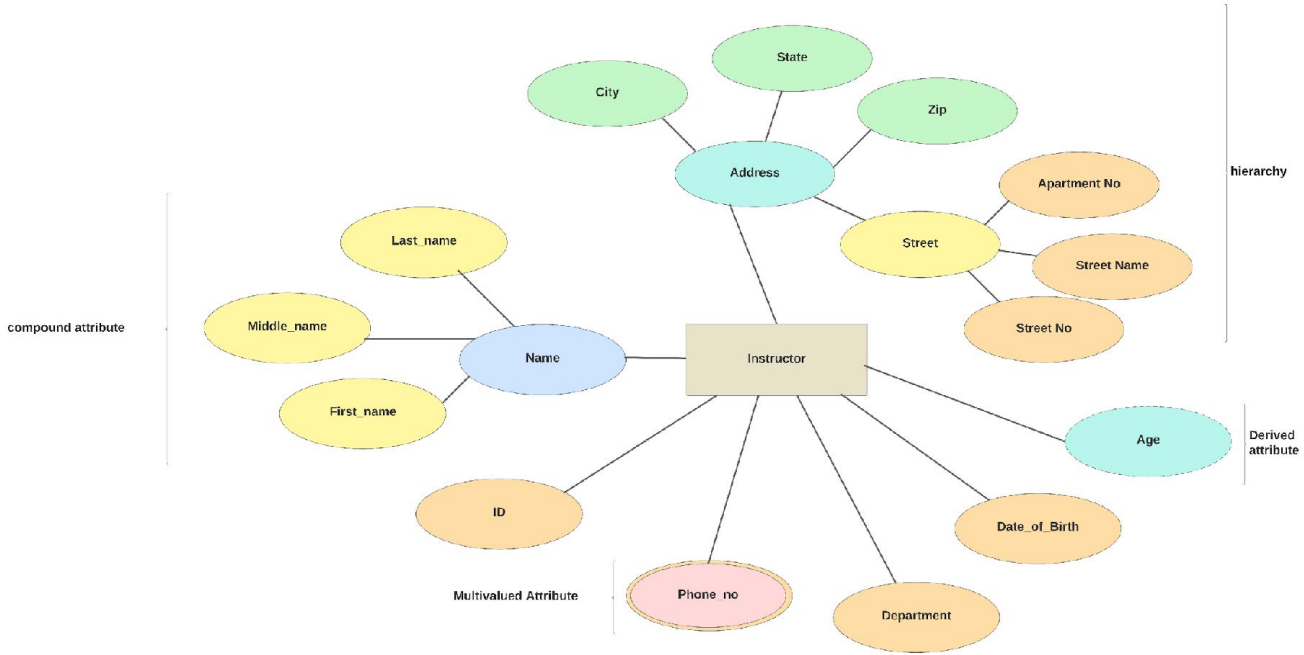
Ternary Relationship Example



Example - 1

- Let us consider the Instructor relation of the university database.
- Every instructor would have a unique ID, along with this the instructor would also have the name, address, phone number, Date of Birth, age, and department details to be there in the database
- Given instructor could be a part of only one department. However, each instructor could have multiple phone numbers.
- The address is subdivided into street, city, state, and zip. The street would be again divided into street_no, street_name, and apt_number.
- The name attribute would be subdivided into first name, middle name, and last name.
- Could you draw an E-R diagram for containing the entity Instructor along with all the attributes?

E-R Diagram


State
City
Zip
Address
hierarchy
Apartment No
Last_name
Street
Street Name
compound attribute
Middle_name
Street No
Name
Instructor
First_name
Age
Derived attribute
ID
Date_of_Birth
Multivalued Attribute
Phone_no
Department

Example - 2

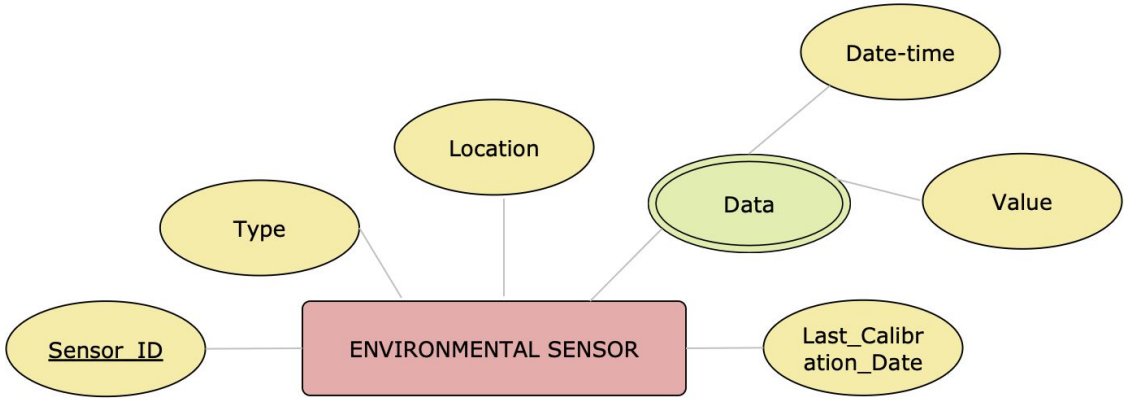
You are designing a database for a **Smart City Infrastructure Management System**. One of the key entities is the **Environmental Sensor**, which is deployed across the city on various infrastructure to collect environmental data like air quality, noise levels, etc.

Model the **Environmental Sensor** entity in an ER diagram with the following details:

- Each sensor must be uniquely **ID**entified.
- Each sensor has a **Type** (e.g., temperature, humidity, pollution).
- Each sensor is installed at a specific **Location**.
- It collects multiple **Data** readings,containing:
  - **Date-Time** of reading
  - **Value** of the reading
- Each sensor also has a **Last Calibration Date**.

Example - 2: ER Diagram

Solution:



Q & A

1) Which of the following is an example of a recursive relationship?

A) Course prerequisites

B) Instructor-student relationship

C) Student-course enrollment

D) Student-advisor relationship

2) What does a binary relationship set involve?

A) One entity set

B) Two entity sets

C) Three entity sets

D) More than three entity sets

Q & A

3) In an E-R diagram, how is a relationship between two entity sets depicted?

A) By a diamond
B) By a rectangle
C) By a line
D) By an oval

4) Which of the following is true about the relationship set 'prereq' in a university database?

A) It connects students to courses
B) It connects instructors to students
C) It connects courses to other courses
D) It connects projects to instructors

Q & A

5) What does a ternary relationship set involve?

A) Two entity sets
B) Three entity sets
C) One entity set
D) No entity sets

6) In an E-R diagram, how is a key attribute represented?

A) Underlined attribute in the oval
B) Underlined attribute in the rectangle
C) Underlined attribute in the diamond
D) Underlined relationship set

Q & A

7) What does an extension of an entity set refer to?

A) The actual collection of entities
B) The unique identifier for each entity
C) The general set of attributes
D) The relationships between entity sets

8) What does an E-R diagram's diamond represent?

A) Entity Set
B) Attribute
C) Relationship
D) Primary Key

Q & A

PES UNIVERSITY
CELEBRATING 50 YEARS

ANSWERS:

1. A
2. B
3. A
4. C
5. B
6. B
7. A
8. C

Weak Entity

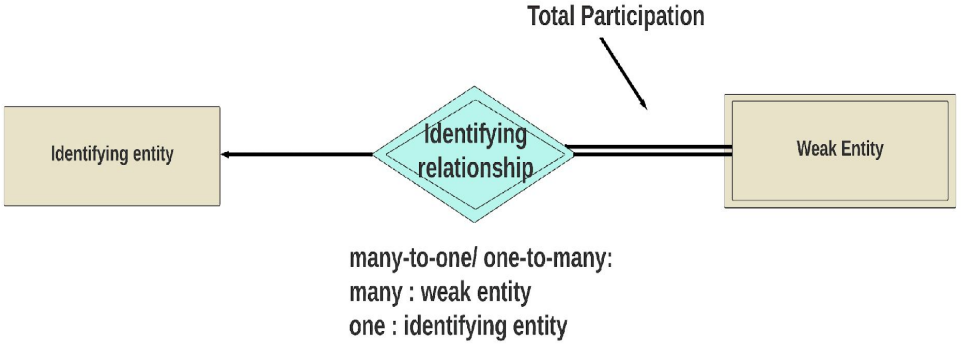
- A weak entity is one whose existence is dependent on another entity, called it's **identifying entity**.
- An entity that is not a weak entity set is termed a **strong entity**.
- Every weak entity must be associated with an identifying entity(Strong Entity); that is, the weak entity set is said to be existence dependent on the identifying entity.
- The relationship associating the weak entity with the identifying entity is called the **identifying relationship**.

Weak Entity

- In the university database, let us consider the following schemas for the entities course and section:
- Course (Course_ID, title, dept_name, credits)
  - Primary key {Course_ID}
- Section (Course_ID, sec_ID, semester, year, building, room_number, time_slot_id)
  - Primary key {Course_ID, sec_ID, semester, year}

Weak Entity

Example:
The ER representation of the weak entity is as shown below



Weak Entity

- For the situation considered for sec_course
  - Weak entity: Section
  - Identifying entity/Strong Entity: Course,
  - Identifying relationship: Sec_course.
  - The key of the section is formed by the key of the identifying entity set (that is, course), plus the discriminator of the weak entity set (that is, section).
  - key is {**course id**, **sec id, year, semester**}.

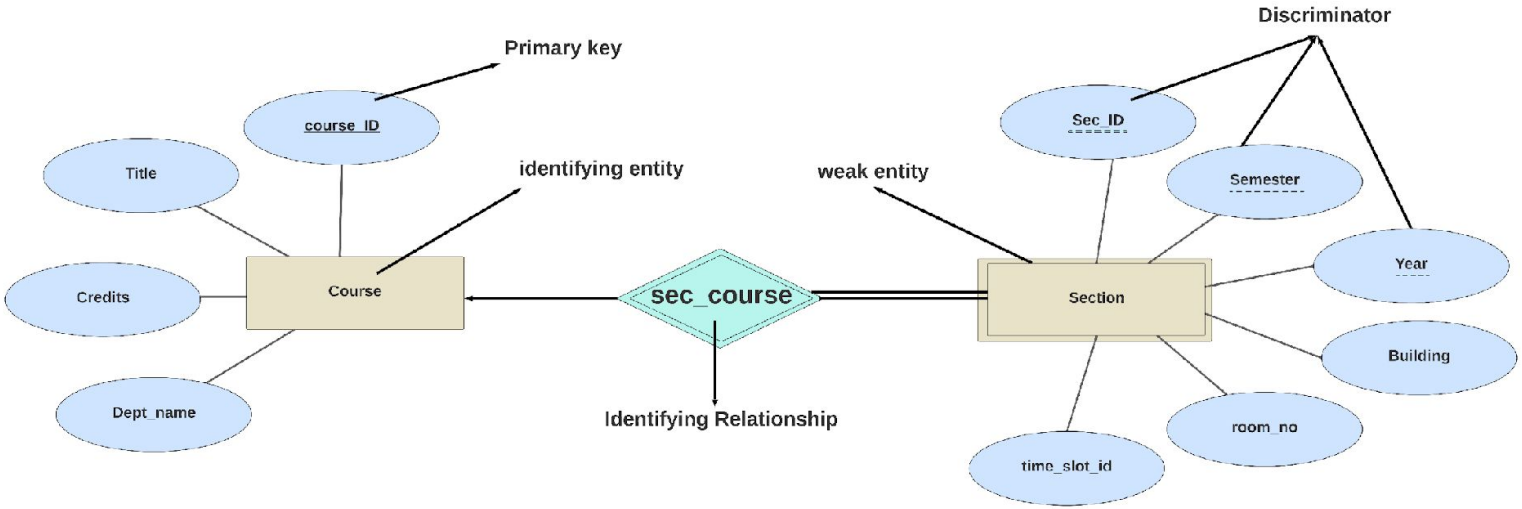
Key(Identifying entity: Course)

Discriminator(weak entity: Section)

Weak Entity

- Note that the ER diagram for the same would be as follows:
  - a weak entity is depicted via a **double rectangle**
  - **discriminator is underlined with a dashed line.**
  - The relationship set connecting the weak entity to the identifying strong entity is **depicted by a double diamond.**

In general, a weak entity set must have total participation in its identifying relationship set



Weak Entity

- **A weak entity can participate in relationships other than the identifying relationship.**

- It is also possible to have a weak entity associated with more than one identifying entity.
- A particular weak entity would then be identified by a combination of entities, one from each identifying entity set.
- The key of the weak entity set would consist of the union of the keys (primary) of the identifying entity sets, plus the discriminator of the weak entity set.

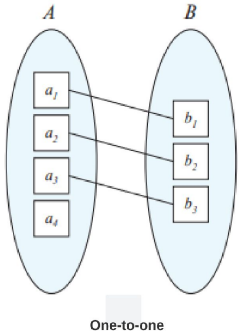
Cardinality Ratio

- Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.
- Mapping cardinalities are most useful in describing binary relationship sets, although they can contribute to the description of relationship sets that involve more than two entity sets.

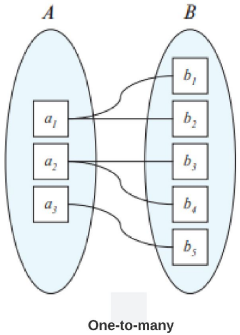
- A binary relationship set R between entity sets A and B, the possible mapping cardinality are:
  - One-to-one
  - One-to-many
  - Many-to-one
  - Many-to-many

Cardinality Ratio

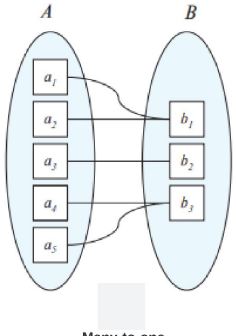
- **One-to-one**
  - An entity in A is associated with at most one entity in B, and an entity in B is associated with at most one entity in A.
- **One-to-many**
  - An entity in A is associated with any number (zero or more) of entities in B. An entity in B, however, can be associated with at most one entity in A.
- **Many-to-one**
  - An entity in A is associated with at most one entity in B. An entity in B, however, can be associated with any number (zero or more) of entities in A.
- **Many-to-many**
  - An entity in A is associated with any number (zero or more) of entities in B, and an entity in B is associated with any number (zero or more) of entities in A.



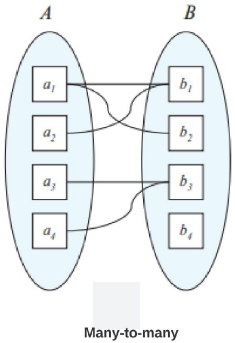
One-to-one



One-to-many



Many-to-one



Many-to-many

Cardinality Ratio

- Let us now consider the advisor relationship set.
- A student can be advised by several instructors - this relationship set can be classified as many-to-many.
- In contrast, if a particular university imposes a constraint that a student can be advised by only one instructor, an instructor can advise several students, then the relationship set from instructor to student must be one-to-many.



one-to-many



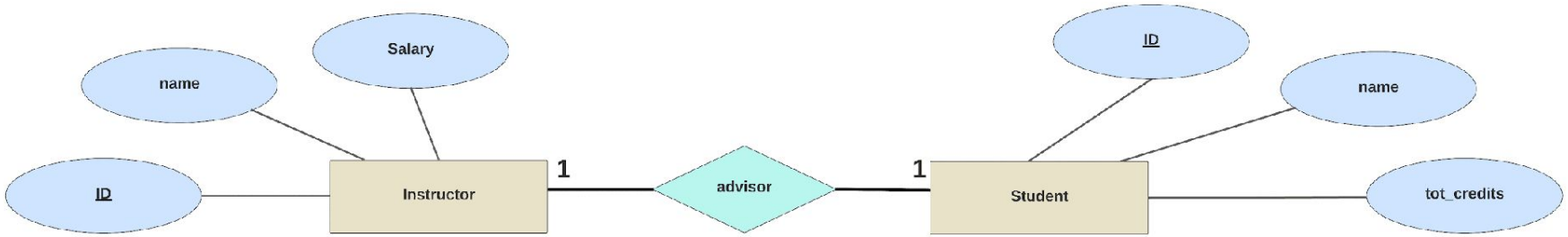
many-to-many

Cardinality Ratio

In the E-R diagram notation, we indicate cardinality constraints on a relationship by drawing an **undirected line (—)** between the relationship and the entity in question.

One-to-one (1:1)

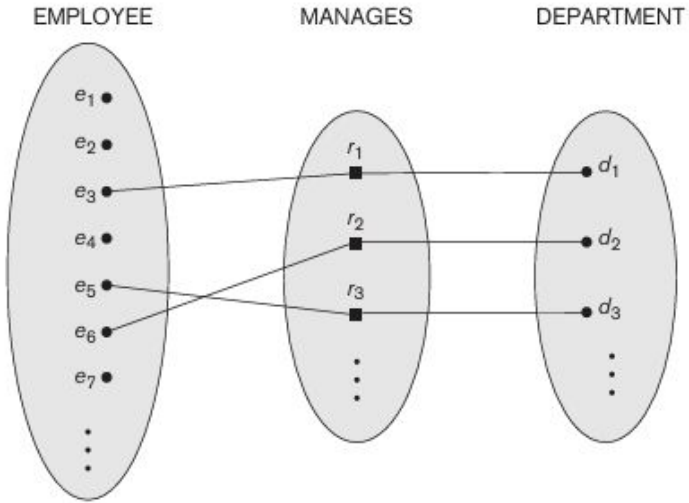
- We draw a line from the relationship set to both entity sets.
- For example, the lines to instructor and student indicate that an instructor may advise at most one student, and a student may have at most one advisor.



Cardinality Ratio

One-to-one (1:1)

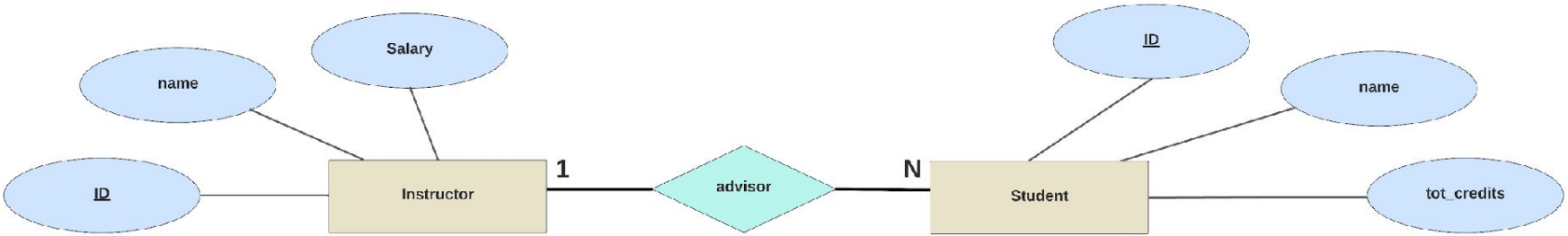
- An example of a 1:1 binary relationship is MANAGES as shown in the figure,which relates a department entity to the employee who manages that department.
- This represents that at any point in time an employee can manage at most one department and a department can have at most one manager.



Cardinality Ratio

One-to-many (1:N)

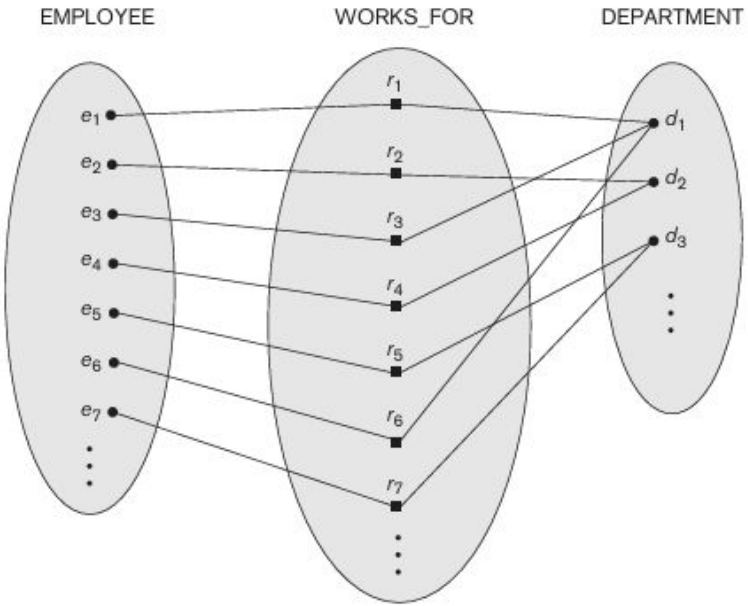
- This indicates that an instructor may advise many students, but a student may have at most one advisor.



Cardinality Ratio

One-to-many (1:N)

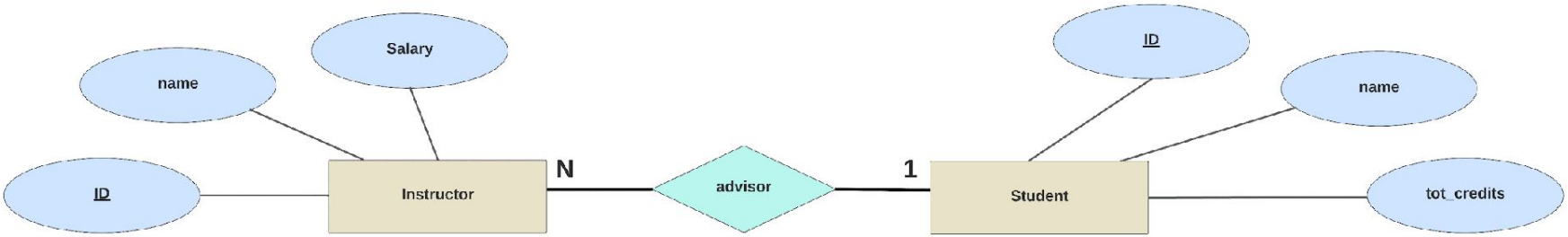
- For example, In the WORKS_FOR binary relationship type, DEPARTMENT:EMPLOYEE is of cardinality ratio 1:N.
- Meaning that each department can be related to any number of employees (N) but an employee can be related to (work for) at most one department.
- This means that for this particular relationship type WORKS_FOR, a particular department entity can be related to any number of employees.



Cardinality Ratio

Many-to-one (N:1)

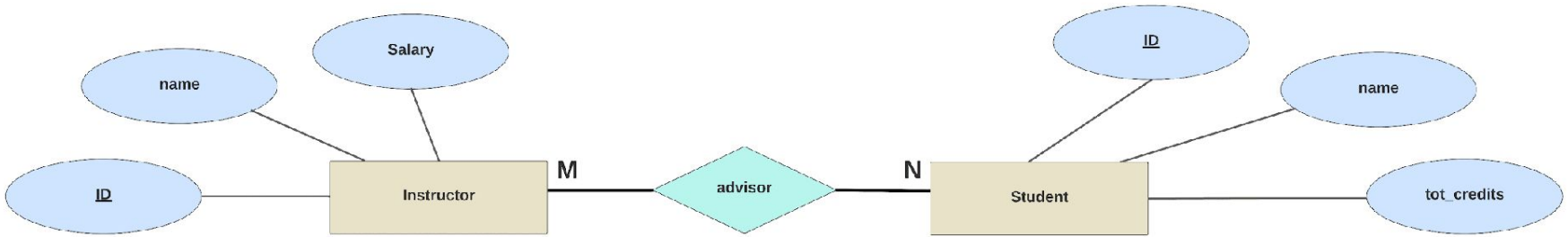
- The converse of the before type
- This indicates that an instructor may advise at most one student, but a student may have many advisors.



Cardinality Ratio

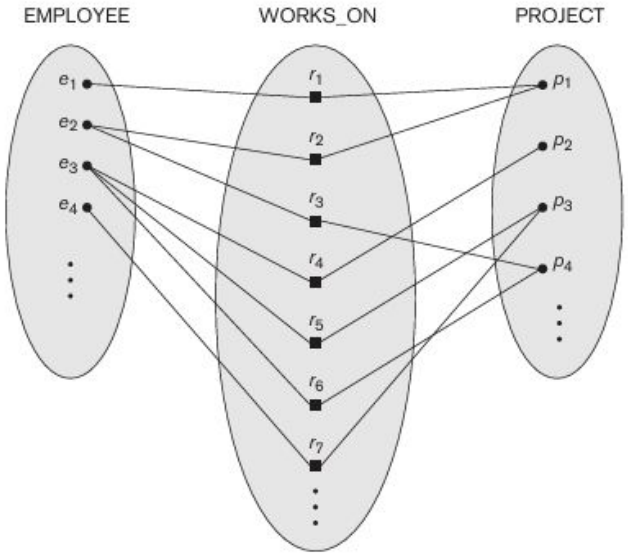
Many-to-many (M:N)

- We draw an undirected line from the relationship set to both entity sets.
- Thus, there are undirected lines from the relationship set advisor to both entity sets instructor and student.
- This indicates that an instructor may advise many students and a student may have many advisors.



Many-to-many (M:N)

- The relationship type WORKS_ON is of cardinality ratio M:N
- The rule is that an employee can work on several projects and a project can have several employees which is represented by the diagram on the right.



Participation Constraints and Existence Dependencies

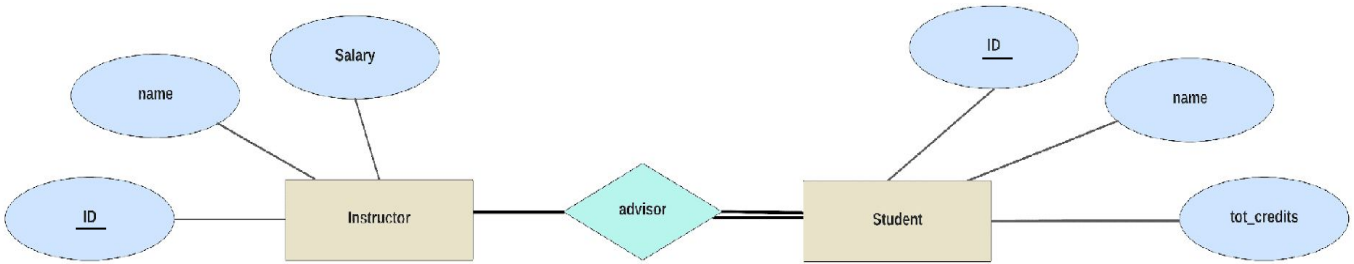
- There are two types of participation :
  - **Total participation**
  - **Partial participation**

- The participation of an entity set E in a relationship set R is said to be **total** if every entity in E must participate in at least one relationship in R.
  - Considering the university database, a university may require every student to have at least one advisor Therefore, the participation of student in the relationship set advisor is total.

- If it is possible that some entities in E do not participate in relationships in R, the participation of entity set E in relationship R is said to be **partial**.
  - An instructor need not advise any students therefore, The participation of instructor in the advisor relationship set is therefore partial

Participation Constraints and Existence Dependencies

- We indicate the total participation of an entity in a relationship set using double lines.
- The example of the advisor relationship set described in the previous slide would look as shown where the double line indicates that a student must have an advisor.



Cardinality ratio

- Let us consider the entities:
  - Student
    - ➢ Attributes: Student_ID, Name, Major
  - Course
    - ➢ Attributes: Course_ID, Course_Name, Department

- The institution has made a rule that given every student has to be enrolled for a minimum of 2 courses and a maximum of 5 courses

- How would we represent this in the ER diagram?

Min-Max notation

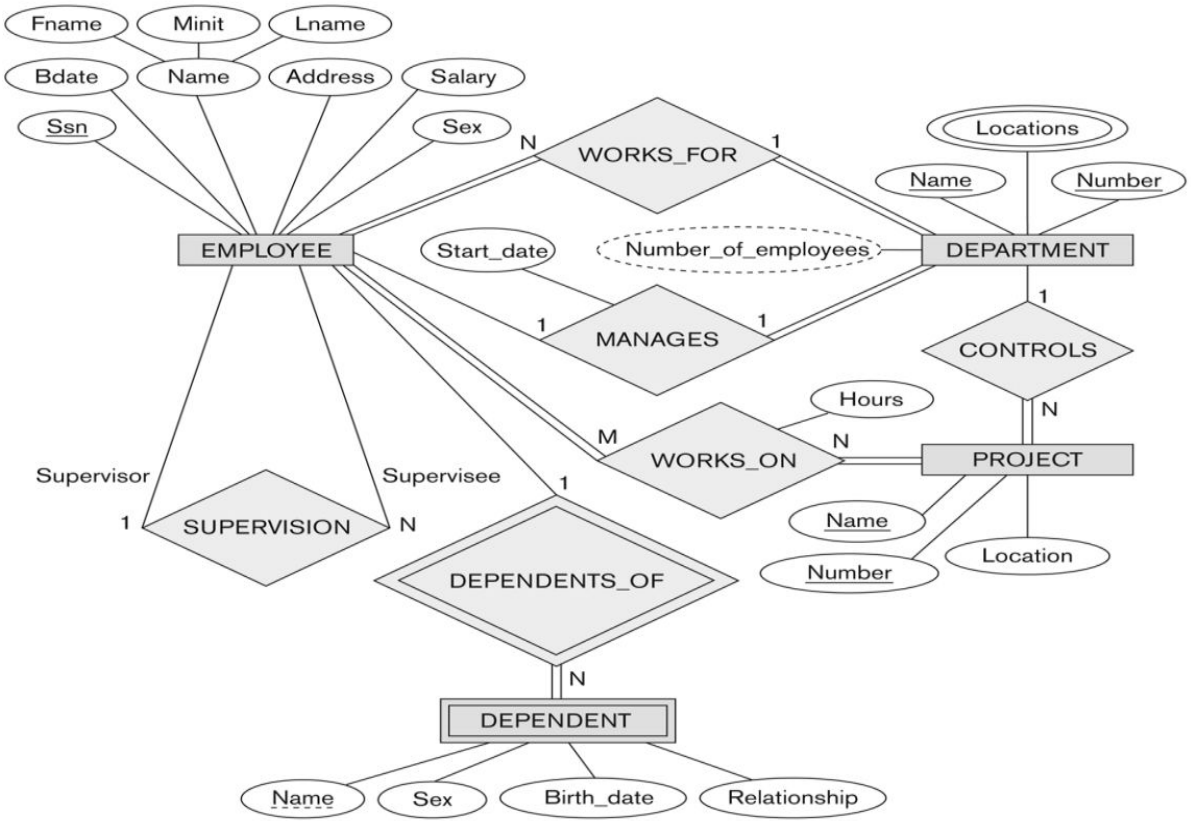
- E-R diagrams also provide a way to indicate more complex constraints on the number of times each entity participates in relationships in a relationship set.
- A line may have an associated minimum and maximum cardinality, shown in the form **(l,h)**
  - **l** is the **minimum cardinality**
  - **h** is the **maximum cardinality**.

- A minimum value = 1 □ total participation of the entity set in the relationship set;
- A maximum value = 1 □ that the entity participates in at most one relationship,
- A maximum value = **N** indicates **no limit**.

Min-Max notation and Chen notation

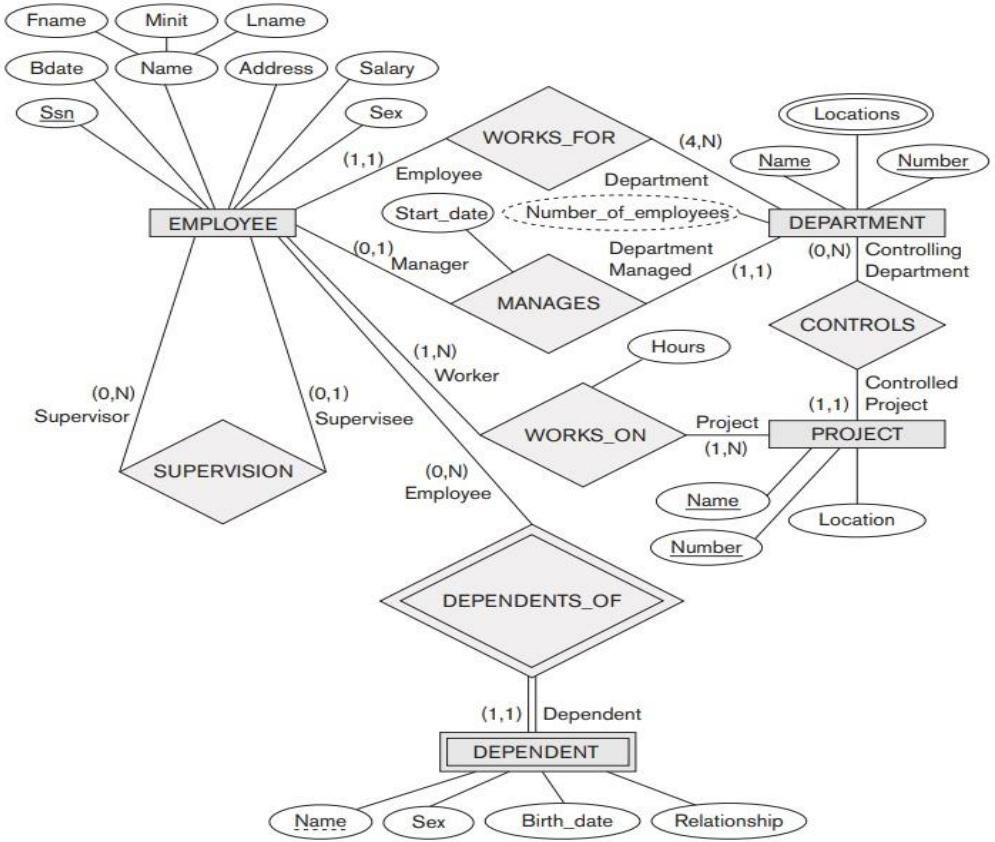
Min-Max [Entity 1]	Chen-Notation	Min-Max [Entity 2]
(0,1)	1:1	(0,1)
(0,N)	1:n	(0,1)
(0,N)	1:n + total part.	(1,1)
(0,N)	n:m	(0,N)
(1,1)	total part. + 1:1	(0,1)
(1,N)	total part. + 1:n	(0,1)
(1,1)	total part. + 1:1 + total part.	(1,1)
(1,N)	total part. + 1:n + total part.	(1,1)
(1,N)	total part. + n:m	(0,N)
(1,N)	total part. + n:m + total part.	(1,N)

Example of Chen Notation



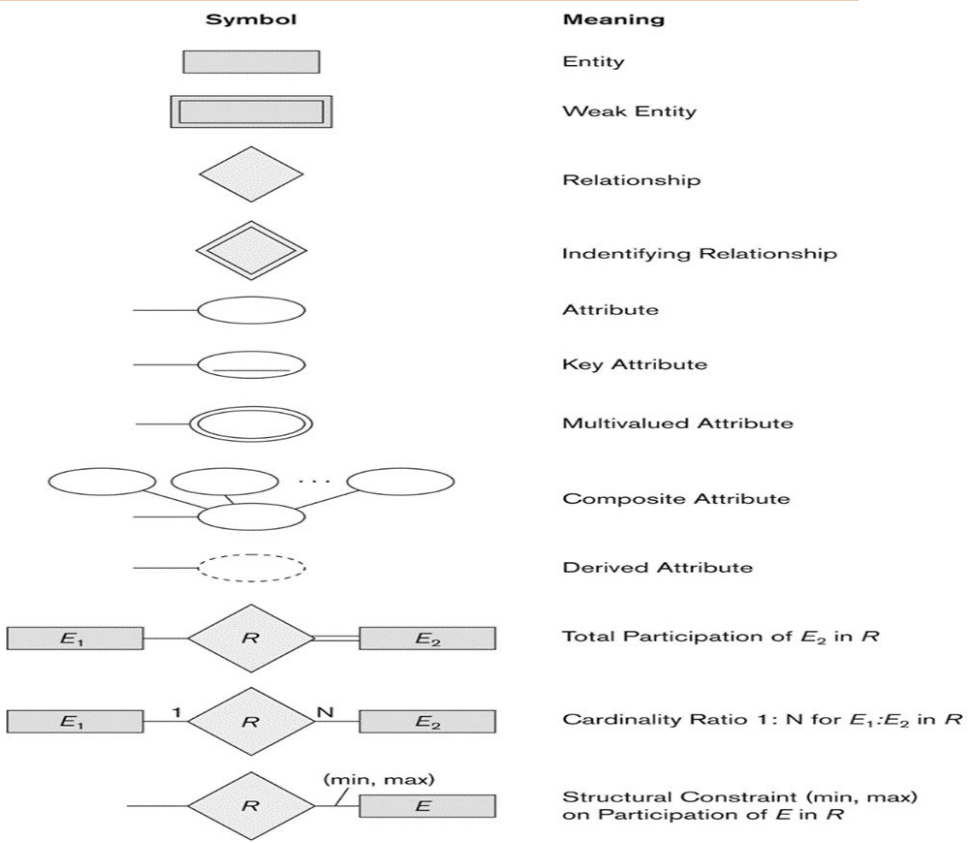
Database Management Systems

(Min,Max) Notation



ER Notations

Figure 3.14
Summary of the notation for ER diagrams.



Example

In the previous slide set, you have already modeled the **ENVIRONMENTAL SENSOR** entity.

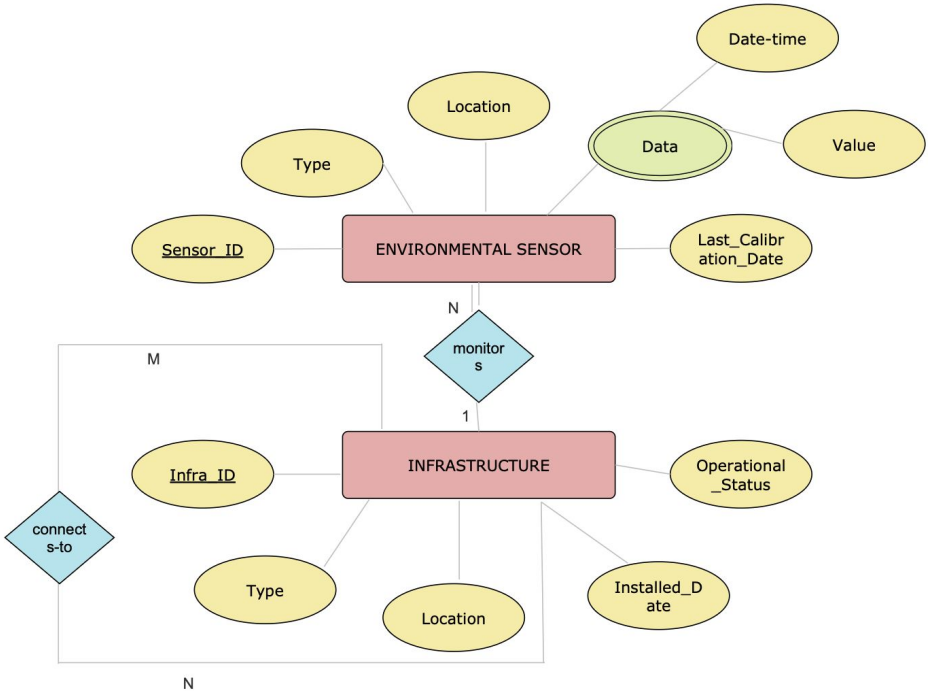
Now, you are provided with additional data related to **urban infrastructure** components. Each **Infrastructure** entity has the following attributes: Infrastructure_ID,Type, Location,Installed_Date and Operational_Status

Based on this:

1. **Draw an ER diagram** that shows the relationship between Environmental Sensor and Infrastructure.
    - Specify the **cardinality and participation (total/partial)** on both sides.
    - Consider that each sensor is installed **on exactly one infrastructure element**, but each infrastructure **may host zero or more sensors**.
2. Identify and model any **recursive relationship** (self-relationship) that may exist within the Infrastructure entity.(Think of how one road may lead to another road).

Database Management Systems

Example



Example

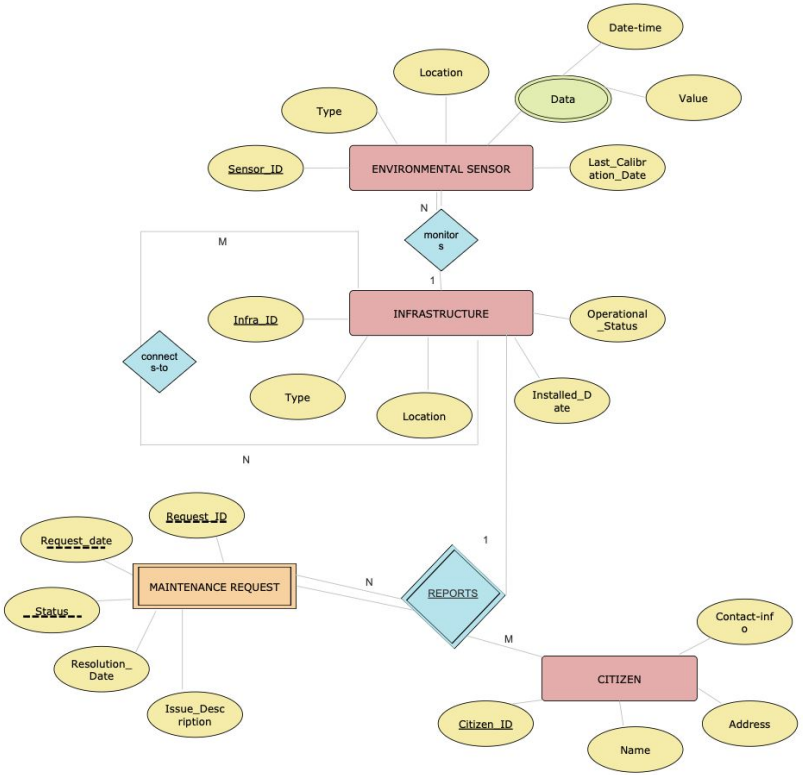
In the same smart urban system, residents often report issues (referred to as **Maintenance Requests)** they observe in public infrastructure—such as broken streetlights, potholes, or malfunctioning traffic signals—via a digital platform. Each request contains detailed specifics like the description, date, and status of the issue.

A single citizen may raise multiple maintenance requests across different infrastructures, and a single maintenance request may be reported by multiple citizens. Each request is always associated with exactly one infrastructure element.However, a maintenance request cannot exist independently—it must always be associated with both a citizen and an infrastructure element.

Task:

- Using this context, extend your existing ER model to include this maintenance request process.
- Be precise in capturing how requests are tracked and related to both citizens and infrastructure.
- Reflect on how the uniqueness of each request might be determined.

Example



In this model, the **Maintenance Request** entity is correctly represented as a **weak entity**, as it depends on both the **Citizen** and the **Infrastructure** entities for its identification and existence.

The attributes associated with the entities are intentionally kept generic to provide flexibility and can be further refined or expanded based on the specific requirements of the implementation context.

With this foundation, you are now well-prepared to **design and interpret Entity-Relationship (ER) models** for real-world systems, ensuring clarity, scalability, and logical consistency in your data modeling practices.

1)Which of the following statements is true about weak entity set?

A. They can exist without being related to a strong entity
B. They always have a multivalued attribute
C. They have a total participation in the identifying relationship
D. They have their own primary key

2)In an ER diagram, a weak entity set is represented by:

A. A rectangle with a double border
B. A diamond with a dashed border
C. An ellipse with a dashed border
D. A rectangle with a dashed border

3) In an ER diagram, the identifying relationship of a weak entity is represented by:

A. A dashed line diamond
B. A double diamond
C. A triangle
D. A rectangle with a double border

4) **Which of the following must a weak entity have?**

A. A discriminator (Partial key)
B. A multivalued attribute
C. A recursive relationship
D. A derived attribute

Primary Key In Relationship Sets

5) **Which of the following best describes why a weak entity cannot exist independently?**

A. It lacks a relationship with any other entity

B. It depends on the foreign key of another entity set

C. It does not have a key attribute of its own

D. It always participates in recursive relationships

6) **In the ER model, a double line connecting an entity to a relationship signifies:**

A. The relationship is one-to-one

B. Partial participation

C. Total participation of the entity in that relationship

D. Recursive relationship

7) A university policy mandates every student must enroll in at least one course, but instructors may or may not teach any course. What participation constraints apply?

A. Total for both Student and Instructor

B. Partial for both

C. Total for Student, Partial for Instructor

D. Partial for Student, Total for Instructor

8) In min-max notation, if the line from Student to Enrolls has (2, 5), what does it represent?

A. A student can enroll in at most 2 and at least 5 courses

B. Each student must enroll in exactly 5 courses

C. A student can enroll in minimum 2 and maximum 5 courses

D. The enrollment is optional

Answers:

1. C
2. A
3. B
4. A
5. C
6. C
7. C
8. C



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

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